

CIS 4374 – Homework 5

Resource and Cost Management

Project: Smart Parking Platform

Team: WE ARE Software Corp.

Submitted by: Muaaz Ismail

This document adds three things to our project: who is on the team and how much of their time we need, what the project will cost and how that money will be spent month by month, and a RACI matrix that shows who does what.

1. Roles and Resources

We planned the MVP of the Smart Parking Platform as a 4-month project (16 weeks). One person working full time is 40 hours a week, which gives 640 hours over the project. Each person's allocation is a share of that. The names are fictional, as the assignment asks, but the roles match the owners we named in our risk register last week.

Role	Name	Skill level	Allocation	Hours	Rate	Cost
Project Manager (also Finance/Billing Lead)	Sara Khan	Senior	25%	160	\$50	\$8,000
Product Owner / Business Analyst	Daniel Reyes	Mid-Senior	20%	128	\$45	\$5,760
Backend/API Lead Developer	Omar Farooq	Senior	75%	480	\$50	\$24,000
Mobile App Developer	Priya Nair	Mid	75%	480	\$42	\$20,160
Payment Integration Lead	Liam Chen	Mid	50%	320	\$42	\$13,440
Mapping/IoT Lead Developer	Ayesha Malik	Mid	50%	320	\$42	\$13,440
UX Designer	Noah Brooks	Mid	25%	160	\$38	\$6,080
QA Lead / Tester	Hannah Lee	Mid	50%	320	\$35	\$11,200
DevOps Engineer	Carlos Mendez	Mid-Senior	20%	128	\$48	\$6,144
TOTAL				2496		\$108,224

Assumptions behind these numbers

- **Skill levels.** Senior means about 6 or more years of experience and can lead design decisions. Mid means about 2 to 5 years and works independently on well-defined tasks.
- **Hourly rates.** These are fully loaded rates, meaning they already include benefits and overhead, so I did not add a separate overhead line.
- **Part-time roles.** The Project Manager, Product Owner, UX Designer and DevOps Engineer are shared with other work, which is why their allocations are low. Developers carry most of the load.
- **Double duty.** The Project Manager also acts as our Finance/Billing Lead, since this is a small team and budget tracking is a small weekly task.
- **Skill gap.** The risk register flagged real-time WebSockets and geolocation as gaps. The senior backend lead covers WebSockets, and we budgeted for a short training workshop for the rest of the team.

2. Resource and Cost Plan

I used a bottom-up estimate, because we know the roles and tasks well enough to count hours for each person, and bottom-up is usually more accurate than a top-down guess. Labor is each person's hours multiplied by their rate. The other costs come from the budget categories in Lecture 6.

Cost plan

Item	Category	Basis	Total	Notes
Team labor	Labor	2,496 hours (see resource table)	\$108,224	Loaded hourly rates
Cloud hosting (AWS)	Infrastructure	\$400/month x 4 months (production + staging)	\$1,600	OPEX
Third-party APIs	Third-party APIs	Maps and SMS, about \$250/month x 4 months	\$1,000	OPEX
Licenses and tools	Licenses & Tools	Jira, GitHub, Figma and dev tools, one-off	\$1,200	CAPEX (one-time)
IoT sensor hardware	Hardware	20 sensors x \$75 + 2 gateways x \$150	\$1,800	CAPEX
Security audit / pen test	Legal & Compliance	One external audit before launch	\$3,000	OPEX
Team training	Training	WebSockets and geolocation workshop	\$800	OPEX
Privacy and payment compliance review	Legal & Compliance	Privacy policy, payment terms review	\$1,000	OPEX
Subtotal			\$118,624	
Contingency (15%)	Reserve	15% of subtotal	\$17,794	Rounded to the nearest dollar
GRAND TOTAL			\$136,418	

Why this contingency, and what is CAPEX vs OPEX

- Lecture 6 says contingency is usually 5 to 20 percent depending on uncertainty. I chose 15 percent because seven of our ten top risks are rated High, including payment API failure, scalability and sensor integration. Contingency covers these known risks. A management reserve for unknown risks would be held by the sponsor and is not in this number.
- One-time CAPEX costs are the sensors and the tool licenses, about \$3,000 in total. Everything else, including salaries, cloud, APIs, the security audit and training, is OPEX.
- As a sanity check against a top-down view, labor is about 79 percent of the total. The lecture example uses roughly 80 percent for labor, so the proportions look reasonable.

Monthly cashflow

Month 1 is planning, design and setup, so labor is lighter and we pay for licenses and training. Months 2 and 3 are the heavy development months, and the sensors arrive in month 2. Month 4 is testing, the security audit and launch. Contingency is spread at 15 percent of each month's spending.

Month	Labor	Cloud + APIs	Licenses	Hardware	Other	Contingency	Total
Month 1	\$22,000	\$650	\$1,200	\$0	\$800	\$3,698	\$28,348
Month 2	\$33,000	\$650	\$0	\$1,800	\$0	\$5,318	\$40,768
Month 3	\$33,000	\$650	\$0	\$0	\$1,000	\$5,197	\$39,847
Month 4	\$20,224	\$650	\$0	\$0	\$3,000	\$3,581	\$27,455
Total	\$108,224	\$2,600	\$1,200	\$1,800	\$4,800	\$17,794	\$136,418

Spending builds up as \$28,348 after month 1, \$69,116 after month 2, \$108,963 after month 3, and \$136,418 at the end. "Other" is training in month 1, compliance review in month 3 and the security audit in month 4.

Hidden costs we are watching

- **Cloud hosting.** Besides compute, we can be charged for storage, backups, monitoring and data transfer out. We will set billing alerts on AWS and tag every resource by project.
- **API usage.** Mapping and SMS calls are billed by volume. For example, at \$0.001 per call, 10 million calls would cost \$10,000, far more than our \$1,000 API line. We will cache map results and add rate limiting to keep calls down.
- **Licenses.** Per-seat tools get more expensive when the team grows, and upgrades can add cost later.
- **Technical debt.** If we rush to hit the deadline, we pay later in refactoring and maintenance. Code reviews and CI checks help keep this under control.
- **Support and security.** Patching, on-call time and compliance (such as payment and privacy rules) continue after launch. That is why we budgeted for a security audit now.

How we will track the budget

- Weekly burn rate and a short variance report (planned vs. actual) from the Project Manager.
- Monthly Earned Value check: the baseline (BAC) before contingency is \$118,624. We compare PV, EV and AC and calculate SPI (EV / PV) and CPI (EV / AC). Anything below 1.0 means we are behind schedule or over budget and we need a response.
- Any change to scope or budget goes through the change control process before it is approved. This also supports our scope creep response from the risk register.

3. RACI Matrix

The matrix below covers the main deliverables of the project. Every row has exactly one Accountable (A) person. Where the same person does the work and approves it, I wrote A/R.

Key: R = Responsible (does the work), A = Accountable (approves it), C = Consulted (two-way input), I = Informed (kept up to date). **Roles:** PM = Project Manager, PO = Product Owner, BE = Backend/API Lead, MOB = Mobile Developer, PAY = Payment Integration Lead, IoT = Mapping/IoT Lead, UX = UX Designer, QA = QA Lead, DevOps = DevOps Engineer.

Deliverable	PM	PO	BE	MOB	PAY	IoT	UX	QA	DevOps
Requirements and user stories	C	A/R	C	C	I	C	C	C	I
UI/UX design and prototypes	I	A	I	C	I	I	R	C	I
System architecture and API design	I	C	A/R	C	C	C	I	I	C
Mobile app development	I	C	C	A/R	C	I	C	I	I
Backend services and database	I	I	A/R	C	C	C	I	I	C
Payment integration (main + fallback)	I	C	C	C	A/R	I	I	C	I
Sensor and mapping integration	I	C	C	C	I	A/R	I	C	C
CI/CD pipeline and cloud setup	I	I	C	C	I	I	I	I	A/R
Test plan and system testing	I	C	C	C	C	C	I	A/R	I
Security audit and fixes	A	I	R	C	C	C	I	C	R
Budget tracking and reporting	A/R	I	I	I	I	I	I	I	C
Release and deployment	A	C	C	C	I	I	I	C	R

A few notes on the matrix

- The Product Owner is accountable for requirements and design because they speak for the city and garage operators.
- Technical leads are accountable for their own areas (backend, mobile, payments, sensors). This keeps decisions close to the people who know the work.
- Security is the one shared item: the PM is accountable, while the backend lead and DevOps do the work.
- QA is Consulted on most integrations so that testing is planned early instead of at the end.